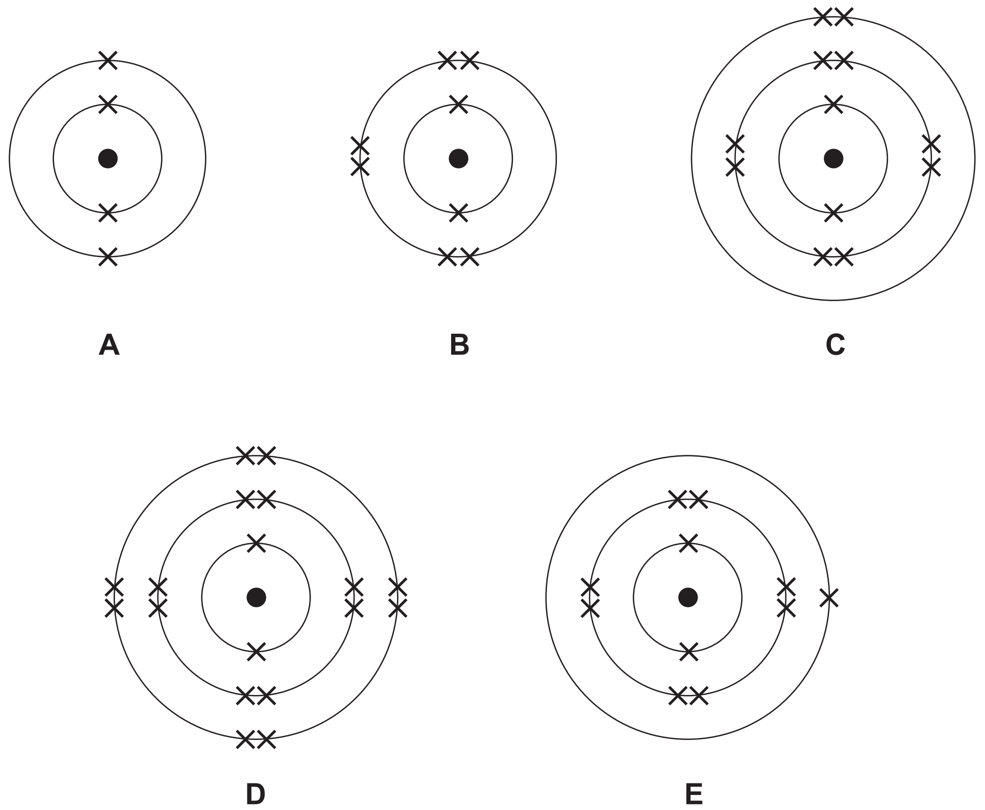
Examiner only

7. The diagrams below show the electronic structures of five elements, **A**, **B**, **C**, **D** and **E**.

The letters are not the symbols of the elements.



(a) Give the **letters** of the elements found in Period 2 of the Periodic Table. Give a reason for your choice. [2]

Letters .... and ....

Reason ....

....

 (3430U20-1)

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(b) Give the **letter** of the element found in Group 0 of the Periodic Table. Give a reason for your choice. [2]

Letter ....

Reason ....

....

(c) Explain how the electronic structure of element **E** can be used to determine its atomic number. [2]

....

....

....

(d) One of the diagrams represents the element oxygen. Oxygen reacts with potassium to form potassium oxide.

Give the formula for potassium oxide and balance the equation for this reaction. [2]

$$\boxed{}\ K + O_2 \longrightarrow 2\ \ldots\ldots\ldots\ldots$$

8